Testing Mailers

Testing Rails mailers involves two separate bits of functionality: specifying whether the email gets sent as a result of some action, and specifying the contents of that email. Specifying whether the email gets sent often starts as part of a request or system test, while specifying the content has a lot in common with view testing. The somewhat indirect nature of the Rails ActionMailer makes testing email less obvious than it might be, but it's not hard. We'll also look at a third-party library that makes email testing easier.

Let's say, for example purposes, that you want to send an email in your project-management system when a task is marked as complete. You don't have a user model yet (I'll talk about users in this system when we get to Chapter 13, *Testing for Security*, on page 261), so let's assume for the moment that all the emails go to some kind of common audit or monitoring address. (Insert `nsa.gov` joke of your choosing.)

At this point you have one of those weird, unique-to-book-examples problems. Specifically, you haven't written very much of the task-tracker site—just a project index and a new-project page. You don't have a list of tasks on a single project, let alone a way to mark a task as complete.

Let's wildly hand-wave our way out of that, writing a request test for just the mail being sent when a task is complete. This will probably imply the existence of pages or functionality you haven't really written yet, but let's all agree to pretend it's already there.

With that hand-waving out of the way, let's write the test, starting with thinking about what you need:

- Given: You'll need one task that starts off incomplete.
- When: The action of this test is a controller action. In a RESTful Rails interface, that action would be `TasksController#update`. Let's go with that. The controller action has a `completed: true` parameter.
- Then: The task updates and an email is sent.

There's a simple case to start with, where `completed` is not set and no email is sent. Writing that test first will let you write the structure of the method.

Prescription 26	When testing a Boolean condition, make sure to write a test for both halves of the condition.